



Better or Worst For Your Workloads: Azure SQL vs. SQL in Fabric

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Tell Ben and William how we did...





Why are you here?

Feature Comparison



Azure SQL Database



SQL Database in Fabric

Show of hands...



How many have seen/used
Azure SQL Database?

How many have seen/used
SQL in Fabric?



DEMO

Agenda

Architecture

HA/DR

Costs & Pricing

Connectivity & Authentication

Q&A

Summary





Architecture

Control all the things...



**Control
Granularity**



SaaS

PaaS

IaaS



Azure SQL Database

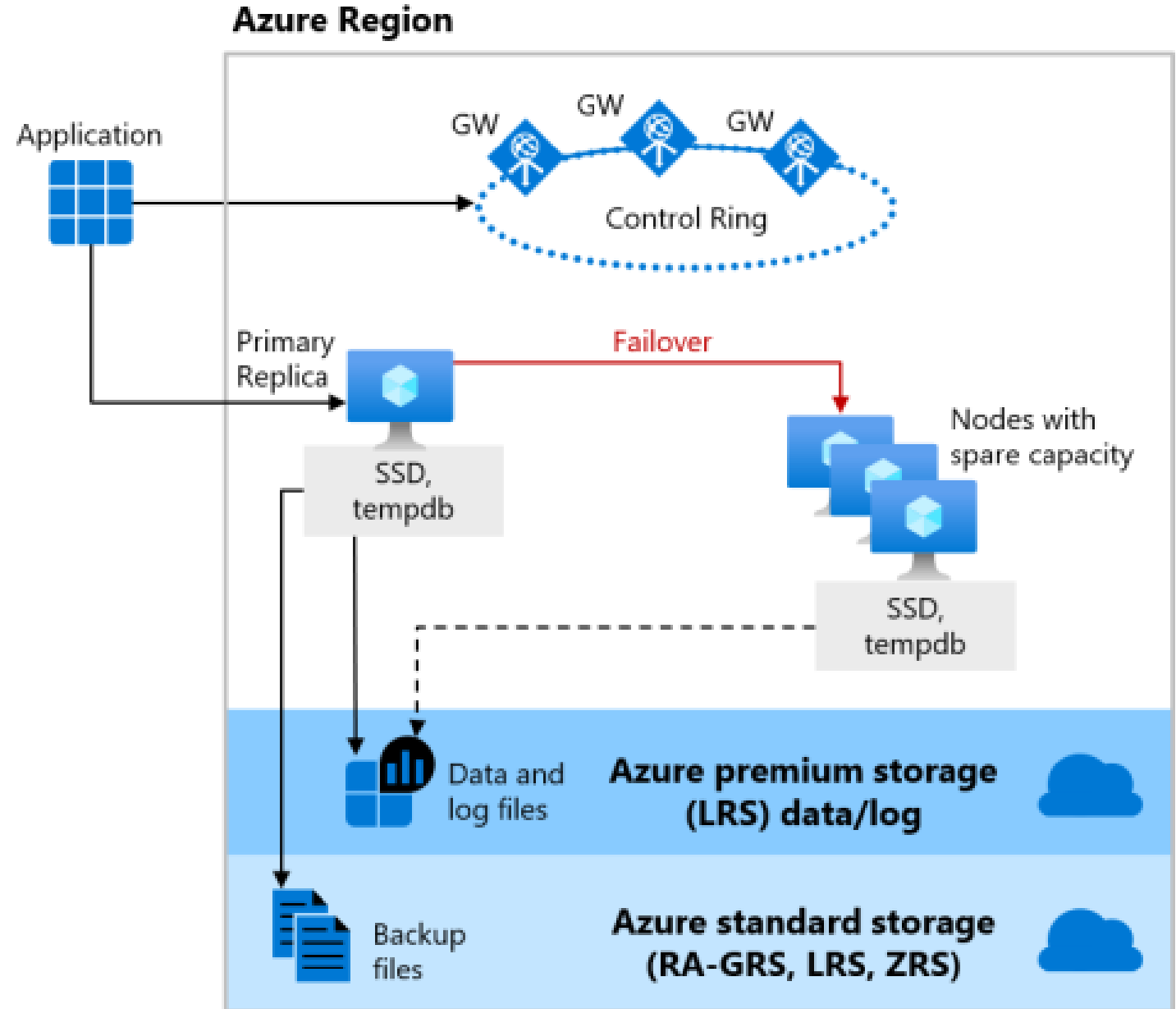


General purpose

Network attached storage

Locally attached SSD for tempdb

Provisioned or
Serverless





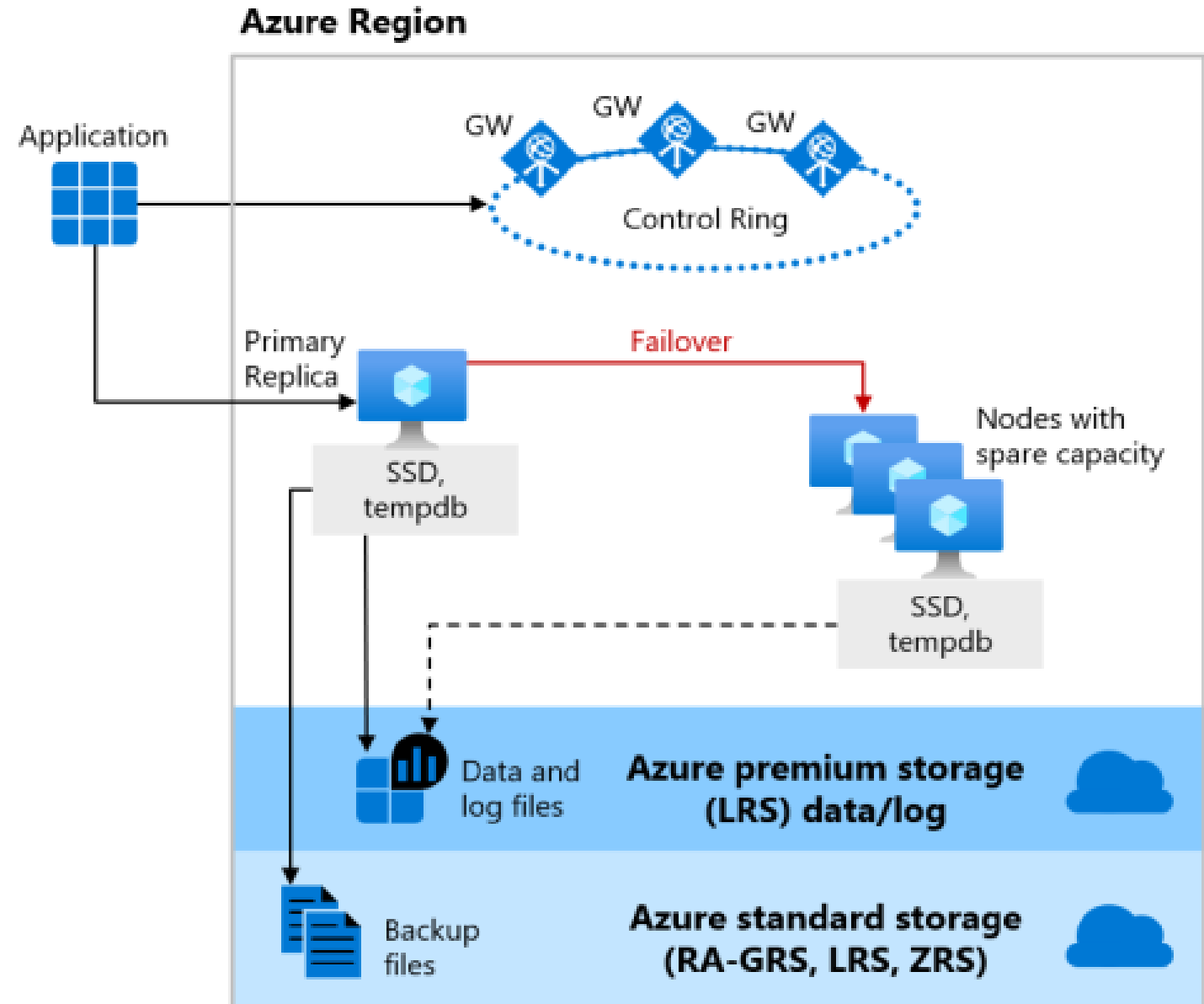
HA/DR

HA/DR



Stateless
compute
layer

Stateful data
layer





Platform feature	Azure SQL Database	Fabric SQL database
Active geo-replication	Yes	No
Automatic backups	Yes	Yes
Availability zones	Yes	Yes, automatically configured
Failover groups	Yes	No
Geo-restore	Yes	No
Long-term retention (LTR)	Yes	No



Development Differences

Development Differences



Feature	Azure SQL Database	Fabric SQL database
Database compatibility	100 - 170	100-170
BULK INSERT statement	Yes, but just from Azure Blob storage as a source.	Yes, via <u>OPENROWSET</u> , with OneLake as data source.
OPENROWSET	Yes, from blob storage	Yes, with <u>OPENROWSET BULK function</u> (preview)
Collation - database collation	SQL_Latin1_General_CP1_CI_AS	SQL_Latin1_General_CP1_CI_AS
TDE	Yes	No
Always Encrypted	Yes	No
Columnstore indexes, clustered	Yes - For certain tiers	Yes, but the index must be created at the same time the table is created, or mirroring must be stopped.



Pricing

Azure SQL DB Pricing



vCore - ~\$184 per provisioned core

vCore Serverless - \$.5218/vCore-hour or \$0.0001450/vCore-second

DTU - ~1.50 per DTU

Elastic pools are available for cost management

Reservations are available

Fabric Capacity



Defined as Capacity Units (CU)

Capacity units is a measurement of time for workloads

Capacity can service multiple workloads

F2 = 2 capacity units, 60 CUs in a 30 second block of time.

1CU = ~0.383 vCores. F64 = ~24 vCores

Reservations are available, elastic pools are not

Capacity Units Pricing»»»

SKU	Cores	Capacity unit (CU)	Pay-as-you-go
F2	0.766	2	\$262.80/month
F4	1.532	4	\$525.60/month
F8	3.064	8	\$1,051.20/month
F16	6.128	16	\$2,102.40/month
F32	12.256	32	\$4,204.80/month
F64	24.512	64	\$8,409.60/month
F128	49.024	128	\$16,819.20/month
F256	98.048	256	\$33,638.40/month
F512	196.096	512	\$67,276.80/month
F1024	392.192	1024	\$134,553.60/month
F2048	784.384	2048	\$269,107.20/month

Pausing Capacity

 **fabricdemo1**   

Fabric Capacity

 Search



 Pause



Refresh



Move



Delete

 Overview

-  Activity log
-  Access control (IAM)
-  Tags
-  Diagnose and solve problems
-  Resource visualizer
- Scale

 Change size
- Settings

 Quick Start
- Capacity administrators

 Essentials

Resource group ([move](#))

[FabricDemo](#)

Status

Active

Location

East US 2

Subscription ([move](#))

[Microsoft Azure Sponsorship - Production](#)

Subscription ID

af11b486-3e53-4077-b62e-f08f0e83ff65

Tags ([edit](#))

[Add tags](#)

Resource name

fabricdemo1

SKU

F2

Fabric Capacity Planning



Primary Capacity— Day-to-day production workloads.
Size appropriately to prevent EOM throttling

Monitoring Capacity—Smaller capacity like an F4, and should just have your monitoring workspace(s) for the Fabric Capacity Metrics app.

Development Capacity—Used for development efforts, minimizes risks of development efforts impacting production workloads.



Connectivity & Authentication

Connectivity & Authentication



Azure SQL DB – SQL Authentication or Microsoft Entra

SQL Server Management Studio (SSMS), Azure Data Studio (ADS) ☒, VS Code, Query Editor (Portal)

Private endpoints

Connectivity & Authentication



SQL Endpoint

Microsoft Entra
Authentication only

VS Code / SSMS

Graphical Query Editor

Managed Private
Endpoints

Visual Studio Code



Copy these strings and use them to connect externally to the SQL database server from client tools. [Learn how to connect to VS Code](#)

Server Name

w6o7roh6pxeeheab3j7nj6yml4-zrny2so5jcwuzk3apmxdx5isem....



Database Name

DB1-d66fcd1d-d9ea-4fa6-a43e-8a008590eea6



Open in VS Code

Download VS Code

Core Feature Comparison



Feature	Azure SQL Database	SQL database in Fabric (preview)
Purchasing models	vCore, DTU	Provisioned Fabric capacity SKU
Compute tiers	Provisioned or serverless	Serverless
Elastic pools	Yes	No
Secure	Azure RBAC, SQL permissions	Fabric workspace roles and sharing permissions, SQL permissions
Authentication	Microsoft Entra ID authentication, SQL authentication	Microsoft Entra ID authentication
Database mirroring to Fabric OneLake	Yes, manually enabled	Yes, automatically enabled for all eligible tables
Cross-platform queries in Fabric	Yes, via mirroring to Fabric	Yes, via Fabric OneLake automatically

Summary



Same basic architecture

SQL in Fabric cannot cross regional boundaries for DR

SaaS vs PaaS (less control)

Connectivity is different

In short...



It's just SQL



So there's no grand prize wiener?



Pick the right wurst, er, database for your workload (that goes for tonight, too)...



References



Overview - <https://learn.microsoft.com/en-us/fabric/database/sql/overview>

Licensing - <https://learn.microsoft.com/en-us/fabric/enterprise/licenses>

Decision Guide - <https://learn.microsoft.com/en-us/fabric/database/sql/decision-guide>

Feature Comparison - <https://learn.microsoft.com/en-us/fabric/database/sql/feature-comparison-sql-database-fabric>



Questions?
Answers!

A person wearing a light-colored suit and a dark tie is holding a white rectangular sign with both hands. The sign has the text 'THANK YOU FOR YOUR ATTENTION' printed on it. The words 'THANK YOU' and 'ATTENTION' are in a dark teal color, while 'FOR YOUR' is in a brownish-orange color. The person's face is not visible, only their hands and torso are shown against a plain, light background.

THANK YOU
FOR YOUR
ATTENTION



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Slides & Demos